Vidya — Product Requirements Document (v0.2, validation-first draft)

Research date: 17 October 2026. Status: DRAFT for founder and engineering review. Scope of this version: sections 1–13 and 18–28 in condensed form, with a prioritized requirements set. Full per-module requirement tables, appendices and a verified competitor matrix are listed as follow-ups in Section 27.

Change log: v0.1 to v0.2

v0.2 narrows this PRD from a feature list to a validation plan. Every priority below is conditional on customer interviews that have not happened yet. Evidence and sources are in the Market Validation Report.

1. **Competitors:** Entab, Fedena and Teachmint checked more deeply; MyClassboard, MyClassCampus, Edunext and Schoollog remain Unverified. Entab customer counts and prices are Conflicting, not resolved.
2. **Corrected, DPDP:** v0.1 relied on secondary sources. The Gazette text (G.S.R. 846(E)) puts most duties, including children's-data rules, eighteen months after publication, about 13 May 2027 (date to confirm). The 72-hour breach duty is now verified (Rule 7). The educational-institution exemption is narrower than v0.1 implied: it covers only tracking and behavioural monitoring for educational activities or safety (Rule 12, Fourth Schedule).
3. **Corrected, stack:** the official Spring Boot 4.1 page says it is compatible up to Java 25. Library-level compatibility and the repository itself remain unchecked.
4. **Downgraded:** v0.1 statements about Teachmint's ERP depth came from a competitor-written page and now carry low weight.
5. **Reprioritised:** see Sections 29 to 34. The P0 and P1 requirement tables in Section 14 are provisional until the Phase 0 gate.

Where Sections 29 to 34 contradict earlier text, they govern; earlier sections will be reconciled in v0.3.

1. Document metadata and evidence caveat

- Owner: Founder / product. Audience: founders, engineering, design, QA, sales.

- Evidence base: about five targeted web searches (market size, DPDP rules, competitor pricing, UDISE+ statistics) plus the architecture you supplied. **This is not exhaustive research.** No customer interviews, Reddit/community complaint mining, or official competitor documentation review has been done yet; those claims are marked *unverified* or *hypothesis*.
- Source quality: most Indian school-ERP comparison content is written by vendors (Entab-style blogs, Fuzen, Devryon, Class ON, Decentro, Codingclave). These contradict each other (see 7) and are treated as low-confidence.

2. Executive summary

Vidya should not launch as an all-in-one school ERP competing on feature count; the field has 20+ vendors and every comparison page claims completeness. The defensible opening is a **reliable, fast-to-onboard core for small and mid-sized private K–12 schools (roughly 300–1,500 students) in CBSE/state-board settings**, with transparent pricing, correct attendance and timetable behavior, and a privacy-by-design parent portal ready for India's DPDP obligations (children's-data rules apply from 14 May 2027, per secondary sources).

The existing tenant domain (attendance, timetable, substitutions, calendar) is a genuine head start. The biggest commercial gaps in the current baseline are **fees, parent/student portals, notifications, exams/report cards and data import**. Those, not more timetable sophistication, decide whether a school can run day-to-day on Vidya.

3. Product overview and vision

- **Vision:** schools run daily operations on one dependable system that staff and families actually use.
- **Customer promise (hypothesis):** a school can be live, with real student data imported and attendance running, within days, at a price known up front.
- **Principles:** reliability before breadth; correctness of money and attendance records; privacy of children's data by default; no hidden add-on pricing; data exportable at any time; incremental evolution of the existing modular monolith.
- **Avoid claiming** "best school ERP", "all-in-one" or "AI-powered" without a demonstrable benefit.

4. Existing system baseline and constraints

Taken as supplied, **not independently verified against the repository**: Java 25, Spring Boot 4.1.1, modular monolith (`com.vidya`), master DB plus one PostgreSQL database per school, `TenantRoutingDataSource` / `TenantContext`, separate master and tenant transaction managers, Flyway (master V20, tenant V16), Redis sessions, stateless JWT (15-min access, 7-day refresh), Testcontainers on PostgreSQL 16.

Open verification items (not done in this draft): confirm Java 25 and Spring Boot 4.1.1 compatibility with every dependency (Spring Session, Lettuce, Hibernate, Flyway, Testcontainers), and confirm Java 25 LTS/support posture before production commitments.

4.1 Known limitations: classification and priority

#	Limitation	Class	Customer impact	Priority
1	One subject/teacher/room per section slot; no electives or split batches	Confirmed product gap	Blocks senior-secondary schools with electives and any language-group split	P1 (P0 if the beachhead includes classes 11–12)
2	Bulk timetable replace regenerates slot IDs	Technical constraint	Breaks any future reference to slots (substitutions, homework, notifications)	P1, fix before building features that reference slots
3	Holiday-aware timetable views incomplete	Confirmed product gap	Teachers see classes on holidays; erodes trust	P1
4	Substitutions use natural-key slot reference, not FK	Technical constraint / risk	Orphaned or wrong substitutions after timetable edits	P1, tied to #2
5	Student/parent portals denied by tenant authorization	Confirmed product gap	No family-facing value, no parent notifications	P0 for Phase 2

#	Limitation	Class	Customer impact	Priority
6	Non-working-day attendance override needs role plus non-blank reason	Existing control, keep	Protects audit integrity	Keep; add reason-quality reporting later (P3)
7	Fifth-Saturday working pattern	Risk requiring investigation	Wrong attendance validation on 5th Saturdays	P1, add explicit tests across year boundaries
8	Asia/Kolkata default timezone, injected shared `Clock`	Technical constraint (good)	Enables deterministic tests	Keep; enforce in review checklist

5. Market landscape and target market

Facts (UDISE+ 2024-25, via secondary analyses of Ministry of Education data): 1,471,473 schools, about 246.9 million enrolments, about 10.1 million teachers. Government schools are about 70% of schools but about half of students; private schools are about 21% of schools yet about 39% of students, and private unaided enrolment is rising. A secondary source reports UDISE+ 2025-26 released 7 July 2026 with about 1.467 million schools; treat as single-source until checked against the official release.

Market-size reports (Technavio/Research and Markets, MarketsandMarkets) are paywalled global estimates with different definitions; Vidya should not use them for planning. No credible India-specific K–12 SaaS market share data was found; vendor claims such as a competitor's 12–15% share are **unverified**.

Hypotheses to validate: private unaided schools are the paying segment; government schools are largely served through state systems and procurement, which is a poor early target; fee collection and parent communication are the strongest purchase triggers.

6. Research methodology and source quality

Source type	Examples	Use	Confidence
Government / official-derived statistics	UDISE+ via NIEPA-linked analysis, Factly/Dataful	Market sizing	Medium-high
Law firm / government-adjacent legal summaries	Mondaq, SCC Online, LiveLaw, EY, Lakshmikumaran & Sridharan, DSK Legal	DPDP obligations	Medium-high; confirm against the Gazette
Review platforms	Capterra, Software Advice, TrustRadius, G2	Pricing hints	Medium, prices vary by listing
Vendor blogs and "best ERP" lists	Fuzen, Devryon, Class ON, Decentro, Sunbeam, Growthjockey, Codingclave	Category structure only	Low
Analyst press pages	Technavio, MarketsandMarkets	Directional only	Low

Vendor-published survey figures (for example an "annual fee leakage" total and push-versus-SMS open-rate claims from a vendor media kit) were **deliberately not used**.

7. Competitor analysis (preliminary, low-to-medium confidence)

Product	Category	Evidence found	Pricing evidence	Confidence
Entab CampusCare	Full school ERP, large/CBSE schools	Widely listed; school counts range from 1,200 to 3,000 across sources	Not published. Estimates conflict: Rs 1.5–8 lakh/yr in one vendor comparison versus Rs 25,000–80,000/yr in another	Low

Product	Category	Evidence found	Pricing evidence	Confidence
Fedena (Foradian)	Full ERP, originally open source	Modules span fees, HR/payroll, library, transport, hostel; review sites mention data-migration effort	Review sites list entry pricing of about $999–$1,299/yr (per feature or plan; listings differ)	Medium
Teachmint	Classroom platform expanding into school management	ERP depth (fees, timetable, HR) described as less mature than Entab/Fedena, per a competitor-authored comparison	Not published; one vendor guesses Rs 100–300/student/yr from community chatter	Low
VidyalayaERP	Cloud school ERP for Indian schools	Gartner listing (updated May 2026) cites a 15-day trial, onboarding support and transparent plans	Not verified	Low-medium
MyClassboard, MyClassCampus, Edunext, Schoollog, Campus 365, Classe365	Various	Appear in lists only	Not verified	Low

Not publicly verified: each product's low-connectivity support, timetable conflict handling, multi-branch consolidation, APIs/webhooks, data-export portability, and customer-complaint patterns. Do not claim Vidya is better on any of these until checked against official docs and independent reviews (action in Section 27).

Observation (not proof): categories in comparison content are open source, freemium mobile-first, small-school, K–12 specialist and communication-first. The "freemium whose paid modules add up" complaint is a recurring vendor-written claim and should be tested in interviews.

8. Customer problems and evidence

All items below are **hypotheses**; prevalence is unmeasured.

Problem	Persona	Evidence status	Existing workaround	Vidya can solve?
Complex fees (installments, concessions, sibling discounts, refunds) and slow recovery	Accountant, owner	Frequently asserted by vendors; no neutral source yet	Spreadsheets, Tally, receipt books	Yes, but needs accounting validation
Manual attendance, corrections without audit	Teacher, admin	Existing Vidya capability addresses this	Paper registers	Yes (exists)
Timetable clashes and last-minute substitutions	Coordinator	Existing capability; electives gap	Manual boards	Partly
School-to-parent communication scattered across WhatsApp	Principal, parent	Plausible; unmeasured	WhatsApp groups	Yes, with consent and cost care
Report cards and exam workflows per board	Coordinator, teacher	Not researched	Excel templates	Needs board-specific research
Onboarding and migration pain	Admin	Fedena review text mentions migration cleanup effort	Vendor-led data entry	Yes, a likely differentiator

Problem	Persona	Evidence status	Existing workaround	Vidya can solve?
Opaque pricing and add-on costs	Owner	Vendor-authored claims only	Negotiation	Yes, if pricing is transparent
Vendor lock-in and poor export	Owner	Not verified	None	Yes, cheap to offer

9. Market gaps and opportunity prioritization

Opportunity	Customer value	Differentiation	Feasibility	Confidence	Validation
Fast, self-service onboarding with validated bulk import	High: time to first value	Medium: copyable but hard to execute well	Medium	Medium	Pilot with 3 schools, measure setup time
Transparent flat pricing, all core modules included	Medium-high	Medium, easily copied	High	Low	Pricing interviews
Reliable attendance plus parent alerts	High	Low-medium	High (attendance exists; needs portal/notifications)	Medium	Pilot
Fee management with a plain-language ledger	High	Low alone, higher if paired with onboarding	Medium-low (money correctness)	Medium	Accountant interviews
Guaranteed data export, no lock-in	Medium	Medium	High	Low	Interviews
Electives/split-batch timetabling	Medium (class 11–12 schools)	Medium	Medium-low	Low-medium	Interview senior-secondary schools

10. Positioning and differentiation (hypothesis)

For owners and principals of 300–1,500-student private schools who lose time to registers and spreadsheets, Vidya is the school operations system that gets you running in days with transparent pricing and records you can trust and take with you. Durable advantages would come from implementation quality, data integrity, and trust, not from features that competitors can copy.

11. ICPs, personas and jobs to be done

Segment	Buyer	Fit	Notes
Small independent (under 300 students)	Owner/principal	Price-sensitive; many alternatives; support-heavy	Low ARPU; likely too costly to serve at launch
Medium private (300–1,500) — recommended beachhead	Owner + principal	Pain is real, budget exists, one site	Best balance; validate in 2–3 states and one board
Large single-campus	Trustees, IT	Needs deeper modules and integrations	Later
Multi-branch groups	Group admin	High value, long sales cycle	Phase 3; master-DB community model suits it
Low-IT schools	Admin	Large overlap with medium segment	Needs hosted onboarding support

Personas (buyer, administrator, daily operator, beneficiary): platform super/ops admin; school owner (buyer); principal/administrator; academic coordinator; accountant; teacher (daily operator); parent (beneficiary, notification recipient); student; school IT/operations staff. Teacher jobs: mark attendance in under a minute, see today's changes, avoid duplicate entry. Parent jobs: know attendance and fee status, receive trusted school notices.

12. Goals, non-goals, principles

- **Goals:** a paying pilot cohort; time to first attendance under one day; zero cross-tenant data incidents.
- **Non-goals for now:** full accounting/ERP, payroll, transport/hostel/library, LMS, AI features, international expansion, government-school procurement.

13. MVP scope and exclusions

Phase 1 (commercially viable core): onboarding and provisioning, school/branch profile, academic year and calendar, student and staff records with CSV import, class/section/subject/teacher assignment, attendance, timetable with fixes to limitations 2–4 and 7, school-level dashboard, audit, authorization hardening.

Phase 2 (daily adoption): fee management, parent/student portals, notifications, report cards for one board, export.

Phase 3: multi-branch analytics, electives/split batches, integrations, selected extra modules.

Excluded from MVP: admissions CRM, library, transport, hostel, payroll, inventory, bus tracking, alumni, AI assistants, public API.

14. Functional requirements (priority set)

Source of each: **E** existing, **N** new, **X** enhancement. MoSCoW: M must, S should, C could, W won't now.

ID	Requirement	Pri	Type	Release
FR-PLT-001	Provision a school tenant database with verified Flyway migration, with recoverable failure states	P0 / M	E→X	MVP
FR-PLT-002	Every tenant request verifies the authenticated user's entitlement to that school; the tenant header alone is never trusted	P0 / M	X	MVP

ID	Requirement	Pri	Type	Release
FR-PLT-003	Subscription state gates features and suspends writes on lapse without deleting data	P0 / M	E→X	MVP
FR-ADM-001	Bulk import of students/staff from CSV/XLSX with row-level validation, dry run and error report	P0 / M	N	MVP
FR-ADM-002	Full tenant data export in open formats on request	P1 / S	N	MVP+
FR-ACA-001	Timetable bulk replace preserves entry identity where slots are unchanged	P1 / M	X	MVP
FR-ACA-002	Substitutions reference timetable entries by a stable identifier	P1 / M	X	MVP
FR-ACA-003	Timetable views mark holidays and non-working days	P1 / S	X	MVP
FR-ACA-004	Elective and split-batch slots (multiple subject/teacher/room per slot)	P1 / S	N	Phase 3 (earlier if beachhead includes classes 11–12)
FR-ATT-001	Bulk daily attendance; non-working-day override requires authorized role and non-blank reason	P0 / M	E	MVP
FR-ATT-002	Fifth-Saturday rule evaluated correctly for configured patterns	P1 / M	X	MVP
FR-ATT-003	Attendance corrections keep an audit trail (who, when, before/after)	P0 / M	X	MVP
FR-FEE-001	Configurable fee heads, class-level fees, installments, concessions, receipts, outstanding balance	P0 / M	N	Phase 2

ID	Requirement	Pri	Type	Release
FR-FEE-002	Refunds/reversals as new ledger entries, never edits of past entries	P0 / M	N	Phase 2
FR-FEE-003	Online payment with idempotent webhook handling and daily reconciliation	P1 / S	N	Phase 2
FR-POR-001	Parent portal showing only that parent's linked children, authorization separate from staff roles	P0 / M	N	Phase 2
FR-NOT-001	Attendance and fee notifications with delivery status, retry, per-guardian preferences	P1 / M	N	Phase 2
FR-EXM-001	Marks entry and report cards for one chosen board's scheme	P1 / S	N	Phase 2
FR-RPT-001	Principal dashboard: today's attendance, absent teachers, substitutions	P1 / S	N	MVP

14.1 Critical acceptance criteria (Given/When/Then)

- **FR-PLT-002.** Given a user authenticated for School A, when they send a request with School B's tenant identifier, then the system rejects it with an authorization error and writes an audit entry, and no School B data is read or written.
- **FR-ATT-001.** Given a school Sunday that is not a working day, when a teacher attempts to mark attendance, then it is rejected; when a school admin submits with a non-blank reason, then it is accepted and the reason is stored.
- **FR-ATT-002.** Given a month with five Saturdays and a pattern that works only on specific Saturdays, when the fifth Saturday is evaluated, then the result matches the configured rule in the school's timezone.
- **FR-FEE-002.** Given a paid receipt, when a refund is recorded, then the original receipt is unchanged, a reversing entry is created, and the outstanding balance recomputes.

- **FR-POR-001.** Given a parent linked to Student X only, when they request Student Y's data, then access is denied regardless of tenant.
- **FR-ACA-001.** Given a timetable with unchanged slots, when a bulk replace is applied, then unchanged slots keep identifiers and existing substitutions remain valid.

15. Critical workflows

1. School onboarding: registration → verification → plan selection → tenant provisioning → import → first attendance.
2. Daily: teacher marks attendance → absent guardians notified (Phase 2) → principal views dashboard.
3. Substitution: teacher absent → coordinator assigns substitute → affected teacher and class see the change.
4. Fees (Phase 2): define structure → assign to students → collect → receipt → reconcile.

16. Role-permission matrix (proposed; verify against current implementation)

Capability	Super/Platform admin	Community admin	School owner	School admin	Teacher	Parent	Student
Tenant provisioning/subscription	Yes	No	No	No	No	No	No
School config, academic year, calendar	Audited support access	Cross-school view	Read	Yes	No	No	No
Student/staff records	Audited support access	Aggregate only	Read	Yes	Own classes read	Own child read	Own read
Attendance	No	Aggregate	Read	Yes plus override	Own classes	Own child read	Own read
Fees	No	Aggregate	Read	Yes	No	Own child	No

Platform and community staff access to tenant student data should be exceptional, time-bound and logged.

17. Non-functional requirements (all targets are proposals to be benchmarked)

Area	Proposed target	Validation
Availability	99.5% monthly at launch, revisit after baseline	Uptime monitoring
API latency	p95 under 500 ms for read endpoints at pilot scale	Load test with realistic school size
Attendance peak	All teachers of a 1,500-student school submitting within a 10-minute window	Performance test
Timetable grid and occupancy	Keep the existing constant SQL-statement count per request, asserted in tests (query count, not total runtime)	Existing invariant tests
Backup/restore	Per-tenant restore tested; RPO and RTO to be set after cost modeling	Quarterly restore drill
Security	Tenant isolation tests for headers, jobs, async, caches, files, exports, webhooks; refresh-token revocation; rate limiting; secure uploads	Authorization and security test suites
Usability	Attendance completed on a phone browser on low bandwidth; Hindi plus one more regional language researched	Usability tests
Accessibility	Aim for WCAG 2.2 AA on parent and teacher flows	Audit

Database-per-tenant operational cost (to model before pricing): connection count scales with active tenants, so a per-tenant pool strategy and idle-pool eviction are required; migrations must be orchestrated across all tenants with failure isolation;

backups and restores are per database; disaster recovery needs automation. This isolation is a strong privacy selling point but raises fixed operating cost per school, so it must be reflected in minimum pricing.

18. Architecture-aware implications

- **Master domain:** identity, school directory, onboarding workflow, tenant registry, subscription, platform billing, audit. **Tenant domain:** all student, staff, academic, attendance, fee and exam data. Do not move student data to master.
- **Consistency:** master and tenant databases cannot share one ACID transaction. Use an explicit workflow with persisted state for provisioning and subscription-to-entitlement changes, idempotent steps, and compensating actions or resumable retries.
- **Payments:** store the gateway event identifier and ignore duplicates; verify signature before any state change; reconcile daily.
- **Tenant context:** set from the authenticated entitlement, propagate explicitly to async jobs, clear in a finally block, include in cache keys and file paths.
- **Portals:** add parent/student authorization via explicit guardian-to-student links inside the tenant domain, not by relaxing the existing denial.
- **Testing:** keep Testcontainers PostgreSQL; add migration tests across tenants, cross-tenant negative tests, and contract tests for webhooks.

19. Security, privacy and compliance

- **DPDP Act 2023 and DPDP Rules 2025** were notified by MeitY on 13–14 November 2025 (sources differ on the exact day; confirm against the Gazette). Per secondary sources, the rollout is phased: Board provisions from notification, consent-manager provisions about 12 months later (14 November 2026), and substantive duties including notice and consent, security safeguards, breach reporting, and verifiable parental consent for children's data from **14 May 2027**.
- Children (under 18) need verifiable parental consent, and tracking, behavioral monitoring and targeted advertising directed at children are restricted. Sources report **exemptions for educational institutions limited to specified purposes**, so whether Vidya is a fiduciary or processor, and which processing falls inside the exemption, is a **legal-review item**.
- Reported breach-notification duties include a 72-hour clock in secondary sources: verify.

- Product requirements: purpose-limited data collection, guardian notice and consent records, retention and erasure workflows, a breach response runbook, processor agreements with each school, and no behavioral profiling of students.
- Needs specialist review: fee receipts and GST/accounting treatment, payment-gateway obligations, board-specific reporting rules.

20. Integrations and dependencies

Payment gateway, SMS/email provider, WhatsApp Business (cost and template policy to research), object storage for documents, accounting export (Tally-style CSV is a likely Phase 2–3 need, unverified). Each external provider needs a fallback and a cost model.

21. Commercial model (hypotheses only; no willingness-to-pay data yet)

Published and rumored competitor figures are inconsistent (Section 7), so treat them as ranges to test, not benchmarks.

Scenario	School	Model	Hypothesis to test
A	About 400 students	Flat annual per school, core modules included	Simple to buy; may under-charge growing schools
B	About 800 students	Per-student per year with a minimum	Scales with value; compare to the rumored Rs 100–300/student range
C	Group of 3+ branches	Per-branch plus group dashboard, annual contract	Higher ARPU, longer cycle

For each scenario, model: hosting per tenant (including the per-database overhead above), payment-gateway fees, SMS/WhatsApp pass-through, onboarding labor, support hours, and break-even number of schools. Charge messaging as usage, and onboarding as a separate fee or bundled into the first year. **Recommended experiment:** show 10–15 principals two price structures (A versus B) and record their stated willingness and objections, then run a paid pilot.

School total cost of ownership includes onboarding time, staff training, devices, SMS and payment fees, and migration effort, not only the subscription.

22. Roadmap and release gates

Phase	Objective	Effort (rough, assumes 2–4 engineers; to be re-estimated)	Go/no-go
0 Discovery	15–20 interviews, 3 pilot schools, pricing test, competitor doc review	4–6 weeks	Three schools commit to pilot
1 Core	Items in Section 13 plus limitation fixes	3–5 months	Pilot schools run attendance and timetable daily for 4 weeks
2 Adoption	Fees, portals, notifications, report cards, export	4–6 months	Fee collection and parent activation targets met in pilot
3 Growth	Multi-branch, electives, integrations	Re-plan after Phase 2	Based on retention

23. Metrics (targets provisional until baselines exist)

Metric	Formula	Source
Time to first attendance	Timestamp of first submitted attendance minus tenant activation	Platform events
Import failure rate	Failed rows / total rows per import	Import logs
Teacher weekly active	Teachers with a write action in 7 days / teachers on roster	Audit log
Parent activation	Guardians who logged in or consented / guardians invited	Portal events
Notification delivery	Delivered / sent per channel	Provider callbacks
Gross margin	(Revenue − hosting, gateway, messaging, support) / revenue	Finance

Metric	Formula	Source
Renewal rate	Renewed schools / schools due	Billing

24. Risk register (top items)

Risk	Likelihood	Impact	Mitigation
Weak differentiation in a crowded market	High	High	Narrow beachhead, onboarding speed, transparent pricing
Fee logic errors damage trust	Medium	High	Append-only ledger, accountant review, extensive tests
Cross-tenant data exposure	Low-medium	Severe	Entitlement checks, negative tests, audit
DPDP non-compliance for children's data	Medium	High	Legal review before the May 2027 obligations apply
Per-tenant database operations cost	Medium	Medium	Cost model, pool eviction, automation
Migration friction blocks adoption	High	High	Import tooling and assisted onboarding
Scope creep into extra modules	High	Medium	Written non-goals and phase gates
Messaging provider cost or policy changes	Medium	Medium	Usage-based pricing, channel fallback

25. Open questions

Blocking MVP: which board and states for the beachhead; whether classes 11–12 are in scope; minimum price that covers per-tenant cost; Vidya's DPDP role versus the school's; Java 25 / Spring Boot 4.1.1 dependency verification. **Deferrable:** WhatsApp strategy, accounting integration format, regional language set, AI use cases, international expansion.

26. Assumption register (starter)

1. Private unaided schools of 300–1,500 students are the paying core. *Validate by interviews.*
2. Fee collection is the top purchase trigger. *Validate.*
3. Schools accept a hosted, per-school database model without extra cost concern. *Validate through pricing.*
4. The supplied architecture description matches the repository. *Verify in code.*

27. Recommendations and next steps

1. **Best initial segment:** medium private K–12 schools (300–1,500 students), single campus, one board, limited IT staff.
2. **Top three problems:** fast reliable onboarding and data import; accurate daily attendance with parent visibility; fee tracking and collection.
3. **First five capabilities:** secure tenant entitlement checks; bulk import; attendance with audit and calendar correctness; timetable fixes and substitutions; principal dashboard. Fees and parent portal immediately after.
4. **Differentiation to test:** transparent pricing combined with a days-not-weeks onboarding promise, backed by exportable data and strong tenant isolation.
5. **Next ten actions:** (1) verify the architecture and dependency compatibility in the repo; (2) interview 15–20 principals, owners and accountants; (3) recruit three pilot schools; (4) check official docs and independent reviews for Entab, Fedena, Teachmint, Vidyalaya, MyClassboard; (5) mine Reddit and school-admin communities for real complaints; (6) obtain legal review on DPDP role and consent design; (7) model per-tenant infrastructure cost; (8) fix timetable limitations 2–4 and 7; (9) design the guardian-linked portal authorization; (10) run the pricing experiment.

Still to produce (not in this draft): full requirement tables per module in the Section 7 format, competitor feature matrix with citations per cell, traceability matrix, glossary and MVP readiness checklist. Ask and I will extend this document.

28. Source bibliography (retrieved 17 October 2026)

- UDISE+ 2024-25 analyses: https://educationforallinindia.com/analysing-distribution-of-udiseplus-2024-25-schools-enrolment-teachers-towards-nep-2020-goals/ ; https://insights.dataful.in/articles/government-schools-account-for-70-of-all-schools-

but-less-than-50-enrolments ; UDISE+ 2025-26 analysis: https://educationforallinindia.com/wp-content/uploads/2026/09/Analysing-Distribution-of-UDISE-2025-26-Schools-Enrolment-and-Teachers-towards-NEP-2020-Goals.pdf (secondary, verify against the official release).

- DPDP Rules 2025: https://www.mondaq.com/regulatory-update-notification-of-the-digital-personal-data-protection-rules-2025/1713500 ; https://www.scconline.com/blog/post/2025/12/26/digital-personal-data-protection-rules-2025-key-highlights/ ; https://dsklegal.com/?p=26455 ; https://anantamias.com/current-affairs/dpdp-rules-2025-notified/?pdf=1 ; https://www.livelaw.in/amp/top-stories/digital-personal-data-protection-rules-2025-notified-309928 ; https://lakshmisri.com/insights/articles/digital-personal-data-protection-rules-2025-incremental-tweaks-and-the-way-ahead (all secondary; confirm against the Gazette notification).
- Competitor and pricing (vendor-authored or listing pages, low-medium confidence): https://www.fuzen.io/posts/school-management-software-comparison-fuzen-vs-entab-fedena-teachmint ; https://codingclave.com/blog/best-school-management-software-india-2026 ; https://www.softwareadvice.com/school-management/fedena-profile/ ; https://us.fitgap.com/products/022107/fedena ; https://www.trustradius.com/compare-products/fedena-vs-teachmint ; https://www.capterra.com/p/175717/CampusCare/ ; https://www.gartner.com/reviews/product/vidyalayaerp-2012477310 ; https://decentro.tech/blog/best-school-erp-software/ ; https://www.growthjockey.com/blogs/school-management-software.
- Market-size press pages (directional only): https://www.technavio.com/report/education-erp-for-private-institutions-market-industry-analysis ; https://marketsandmarkets.com/PressReleases/education-erp.asp ; https://www.researchandmarkets.com/reports/6227622/education-erp-private-institutions-market.

29. v0.2 priority reassessment

Scoring model. Each capability is judged High, Medium or Low on eight criteria: customer impact, evidence strength, differentiation, revenue or retention effect, effort, technical risk, dependencies and operating cost. Customer impact and evidence strength count most. There are no numeric totals, because the inputs are judgments, not data.

Capability	Customer impact	Evidence	Differentiation	Effort and risk	Tier	Note
Tenant authorization and entitlement hardening	High	High (internal need; DPDP Rule 6)	Medium	Medium	P0	No school asks for it, but one cross-tenant leak ends the business
Onboarding and bulk import	High	Medium (Reported, unvalidated)	Medium-high	Medium	P0, conditional	Drops to P1 if interviews do not confirm migration pain
Attendance with audit and calendar fixes	Medium-high	Medium (table stakes)	Low	Low (exists)	P0	Existing; fix fifth-Saturday rules and audit trail only
Timetable reliability and substitutions	Medium	Low-medium	Low-medium	Medium	P1	Fix slot-ID and substitution-reference issues when a feature needs them
Principal dashboard	Medium	Low	Low	Low	P1	Design from interview answers
Fee management	High	Medium (consistent across vendors, all vendor-sourced)	Low unless a recovery workflow proves out	High	P0 for paid launch, not for design-partner pilot	Money correctness needs accountant review
Parent and student portals	Medium-high	Medium	Low	High (consent design)	P1	Start with attendance and fee notifications; legal review first

P2: multi-branch analytics, data export, electives (P1 if classes 11–12 are in scope). **P3:** configurable report templates, partner API, AI features.

Do not build yet: admissions CRM, library, transport and bus tracking, hostel, payroll, inventory, alumni, learning-management or online classes, AI assistants, public API.

30. Known limitations, reclassified

Limitation	Class	Reason
No electives or split batches	Important enhancement; becomes a confirmed MVP blocker only if the beachhead includes classes 11–12	Decide from interviews
Bulk timetable replace regenerates slot IDs	Important enhancement	Fix before features that reference slots
Holiday-aware timetable views incomplete	Important enhancement	Visible to teachers daily
Substitutions use a natural-key slot reference	Requires investigation	Test what happens to substitutions after a timetable edit
Student and parent access denied	Future capability for the pilot; a confirmed blocker for any parent-facing launch	Needs guardian-to-student links and consent design
Non-working-day override needs role and reason	Existing control, keep	Not a limitation
Fifth-Saturday pattern	Requires investigation	Add tests across month and year boundaries; blocker if they fail
Asia/Kolkata default and injected Clock	Not a limitation, keep	Enables deterministic tests

31. Assumption register

ID	Assumption	Evidence	Confidence	Validation	Decide by
A-01	Medium single-campus private schools (300–1,500 students) are the right first segment	Author judgment only	Low	8–12 interviews including branch groups	End of Phase 0

ID	Assumption	Evidence	Confidence	Validation	Decide by
A-02	Fee handling is the top purchase trigger	Vendor claims only	Low	Interview questions 1 and 11	End of Phase 0
A-03	Schools will pay more than a free tier costs	Free plan reported at Teachmint; prices span Rs 0–1,500 per student per year	Low	Pricing experiment	Before any price is published
A-04	Migration pain is large enough to differentiate	Reported for Fedena	Low	Ask about the last migration; measure time to first attendance in pilot	End of Phase 0
A-05	The DPDP deadline makes schools value processor controls	Deadline verified; demand not	Low-medium	Interview question 12; legal review	Before the portal is built
A-06	Per-tenant database cost fits the target price	None	Unknown	Cost model at 50 and 200 tenants	Before pricing
A-07	Java 25 and Spring Boot 4.1.x build cleanly with every dependency	Official page says compatible to Java 25	Medium	Build and CI run on the repository	This sprint
A-08	The supplied architecture matches the repository	Supplied description only	Unknown	Code review	This sprint
A-09	Existing attendance and timetable meet teachers' needs	None	Unknown	Pilot usability tests	During pilot
A-10	Parents will use a portal or app rather than WhatsApp	None	Unknown	Interviews; pilot activation rate	Before the portal is built
A-11	Schools switch systems around the academic-year start	Assumption	Unknown	Interview question 8	End of Phase 0
A-12	The educational-institution exemption covers routine student-data processing	Partial reading of Rule 12	Low	Legal review	Before the portal is built

32. Technical readiness checklist

- [ ] Build and run CI on Java 25 and Spring Boot 4.1.x; compare Spring Session, Lettuce, Hibernate, Flyway and Testcontainers versions with the Boot dependency set
- [ ] Apply master V20 and tenant V16 migrations to fresh and existing tenants
- [ ] Tenant provisioning is resumable and idempotent, with persisted state and a failure-injection test
- [ ] Model connection-pool and database-count limits at 50 and 200 tenants
- [ ] Cross-tenant tests: forged tenant header, async jobs, caches, files, exports
- [ ] Bulk import with dry run, row-level errors and idempotent retry
- [ ] Attendance peak load and the timetable query-count invariant
- [ ] Audit log covers privileged cross-tenant access; logs kept at least one year
- [ ] Monitoring, alerting, per-tenant backup and a tested restore
- [ ] No cross-database transaction assumed; compensation defined for provisioning and subscription changes; payment webhooks idempotent

33. What v0.2 deliberately does not contain

Full per-module requirement tables, the traceability matrix, a complete competitor matrix, the glossary and the MVP readiness checklist are deferred until the Phase 0 gate. Writing them now would harden guesses. The v0.1 risk register still stands. The interview plan and go/no-go thresholds are in the Market Validation Report.

34. Conclusions

1. **Segment:** working hypothesis is medium single-campus private K–12 schools; multi-branch groups are tested alongside. Not confirmed.
2. **Highest-value problems (hypotheses):** getting data in and live quickly; fee tracking and follow-up; trustworthy, compliant handling of children's data.
3. **First five capabilities:** tenant authorization hardening; onboarding and bulk import; attendance with audit; fee management for paid launch; parent notifications, after consent review.
4. **Differentiation:** DPDP-ready, isolated and auditable data handling combined with assisted migration. The deadline is verified; demand is not.
5. **Next ten actions:**

a. Run 8–12 interviews using the plan in the report.
b. Recruit three design-partner schools.
c. Build the per-tenant cost model.
d. Get legal review on fiduciary versus processor and on the exemption.
e. Verify the repository and the dependency build.
f. Check Entab, Fedena and Teachmint pricing pages and demos directly, and research the four Unverified competitors.
g. Search school-administrator communities again for real complaints.
h. Add fifth-Saturday tests and investigate the substitution reference.
i. Design import tooling with dry run.
j. Run the pricing experiment with a small paid pilot.

6. **Assumptions to validate before committing to the MVP:** A-01 to A-06 and A-12.